

App Development-|| Project Report

App Dev 2 Project Report

Student Details

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About Me: I am a student at IIT Madras BS Degree program with a deep interest in web application development. I enjoy building real-world applications that solve complex problems, focusing on clean UI/UX and strong backend logic.

Project Details

Project Title: Campus Placement Portal

Problem Statement:

To design and build a secure, role-based web application that connects Students, Companies, and the Admin. The system streamlines job postings, student applications, interview scheduling, and placement tracking.

Approach:

The app is built with a decoupled architecture. It uses Flask for the backend API and Vue.js for a dynamic frontend. Background tasks like email notifications and CSV exports are handled asynchronously using Celery and Redis to ensure the main website remains fast .

AI/LLM Declaration

I used an AI assistant (Gemini) to debug some limited brainstorm errors and understand Vue.js routes. The core project structure, database architecture, and final implementations were done manually by me.

Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework for building APIs
SQLAlchemy & SQLite	Database management and object-relational mapping
Celery & Redis	Background task queue for async jobs and alarms
Flask-JWT-Extended	Secure role-based user authentication and login
smtplib	Sending automated HTML emails and interview reminders
Bootstrap 5 & Custom CSS	Frontend styling, responsive design, and glassmorphism effects

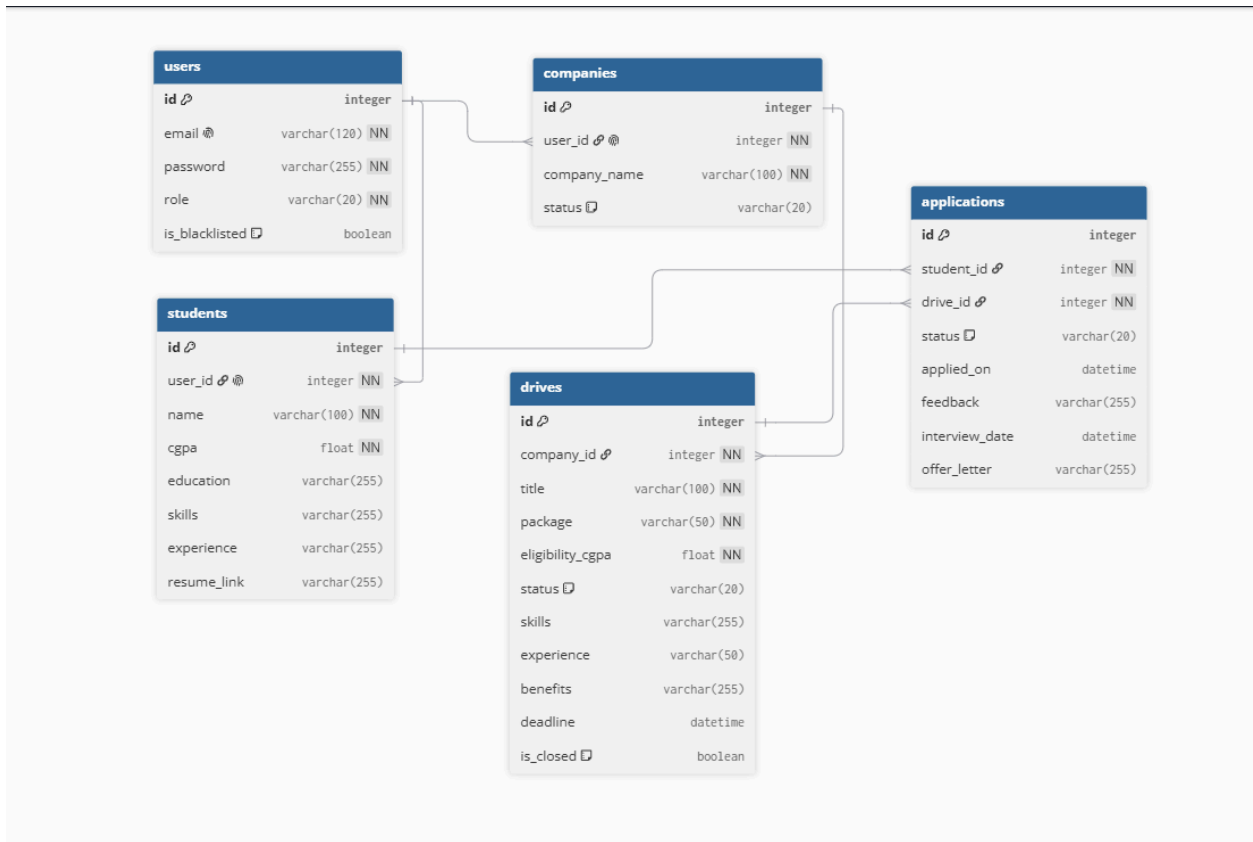
Database Schema / ER Diagram

Tables:

1. **User:** Stores login credentials and roles (id, email, password, role, is_blacklisted).
2. **Student:** Stores student profiles (id, user_id, name, cgpa, skills, resume_link).
3. **Company:** Stores company details and status (id, user_id, company_name, status).
4. **Drive:** Job postings created by companies (id, company_id, title, package, deadline).
5. **Application:** Links students to drives (id, student_id, drive_id, status, interview_date).

Relationships:

- One-to-One → User → Student, User → Company
- One-to-Many → Company → Drive, Student → Application, Drive → Application



6. API Resource Endpoints

Endpoint	Method	Description
/api/auth/login	POST	Authenticate user and generate JWT token
/api/admin/drives/pending	GET	Fetch all pending job drives for Admin approval
/api/company/drives	POST	Create a new job drive by the company
/api/student/applications	GET	Fetch all applied jobs for the student dashboard
/api/company/export	POST	Trigger Celery background task for CSV export

Architecture and Features

Architecture Overview:

- **app.py** – Main Flask application entry point (Backend API).
- **models.py** – Database tables and relationships using SQLAlchemy.
- **/routes** – Flask Blueprints for handling API requests (Admin, Company, and Student).
- **config.py / cache.py** – Configuration for Redis cache and Celery message broker.
- **tasks.py** – Celery worker file for handling background batch jobs (CSV exports).
- **/frontend** – Vue.js application containing all UI components and views.

Implemented Features:

Role-Based Access: Secure login and separate dashboards for Admin, Company, and Student using JWT tokens.

Job Management: Companies can post new placement drives, and Admins can approve or reject them.

Smart Application Tracking: Students can apply for active jobs, and the system automatically prevents double applications.

Application Processing: Companies can update student statuses (e.g., Shortlisted, Interview, Placed) and provide feedback or offer letter links.

Background Batch Jobs: Integrated Celery and Redis to generate large CSV reports in the background without freezing the website.

Auto-Download System: Smart polling mechanism in the frontend that automatically downloads the CSV file once the backend completes the batch job.

Responsive UI: Clean, modern, and easy-to-use interface built for all devices.

Video Presentation

Drive Link:

https://drive.google.com/file/d/1OQCn3t_cUJZZMoRiFopCWPAc5lb9eWu7/view?usp=sharing
